

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
ICSE CLASS 8 - PHYSICS

Chapter: Light

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which mirror is used as a rear-view mirror in vehicles because it gives a wider field of view?

- | | |
|-------------------|--------------------|
| (i) Plane mirror | (j) Concave mirror |
| (k) Convex mirror | (l) Any mirror |

Answer: _____

2. A dentist uses a mirror to see a magnified upright image of a tooth. The mirror is:

- | | |
|------------|-----------------|
| (i) Plane | (j) Concave |
| (k) Convex | (l) Glass plate |

Answer: _____

3. An object is 30 cm in front of a plane mirror. The distance between the object and its image is:

- | | |
|-----------|-----------|
| (i) 30 cm | (j) 45 cm |
| (k) 60 cm | (l) 90 cm |

Answer: _____

4. A ray makes an angle of 30 degrees with the surface of a plane mirror. Its angle of reflection is:

- | | |
|----------------|----------------|
| (i) 30 degrees | (j) 45 degrees |
| (k) 60 degrees | (l) 90 degrees |

Answer: _____

5. A concave mirror forms a virtual, erect and enlarged image when the object is placed:

- | | |
|----------|---------------------|
| (i) At C | (j) Between C and F |
| (k) At F | (l) Between F and P |

Answer: _____

6. For a concave mirror, an object placed at C forms an image that is:

- | | |
|--------------------------|----------------------------|
| (i) Virtual and enlarged | (j) Real and same size |
| (k) Real and diminished | (l) Virtual and diminished |

Answer: _____

7. When white light passes through a prism, the splitting of white light into colours is called:

- | | |
|-----------------------|----------------|
| (i) Reflection | (j) Dispersion |
| (k) Lateral inversion | (l) Diffusion |

Answer: _____

8. The left-right reversal seen in a plane mirror is called:

- | | |
|-----------------------|-------------------|
| (i) Refraction | (j) Dispersion |
| (k) Lateral inversion | (l) Magnification |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. State the two laws of reflection. [2]

10. Write any four properties of the image formed by a plane mirror. [2]

11. For a concave mirror, state the nature and size of the image when the object is (i) at C and (ii) between F and P. [2]

12. Give one use each of a concave mirror and a convex mirror, with a reason for the choice. [2]

13. What is dispersion of light? Write the sequence of colours obtained from white light through a prism. [2]

14. A boy stands 25 cm in front of a plane mirror. How far is his image from the mirror, and what is the distance between the boy and his image? [2]
